09/26

Monday, 25 September 2023 9:15 PM

Review

+ Stress tensor, viscosity & static fluid problems

+ Accounts for intermolecular forces during fluid motion
+ requires mag., direction & orientation

$$\text{Viscosity } (\mu) = \frac{\boxed{\text{shear stress}}}{\text{rate of shear strain}} \sim \text{rate of deformation} \rightarrow \frac{\Delta v}{\Delta x} \sim \frac{dv}{dx}$$

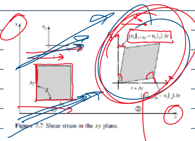


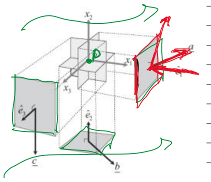
Figure 7.7 Shear strain in the xy plane.

Deformation occuring in two directions due to fluid flow

$$\underbrace{-\frac{dS_{xy}}{dt}}_{\text{deformation}} = \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}$$

$$\boxed{\text{Shear stress}} = \underset{\mu}{(\text{viscosity})}\ \underset{\left(\frac{-dS}{dt}\right)}{(\text{rate of shear strain})}$$

$\boxed{\frac{dv}{dy}}$ or $\dot{\gamma}_0$



$\boxed{xy, yz, xz}$

Similar analysis on each plane } For stress at a point P in fluid motion

$\boxed{(xy, yz, xz)}$

$\frac{\partial v}{\partial y}$ $\frac{\partial v}{\partial x}$ $\frac{\partial v}{\partial y}$ $\frac{\partial v}{\partial z}$ $\frac{\partial}{\partial x}$ $\frac{\partial}{\partial z}$

+

Normal stresses ✓

gives 9 elements to describe all forces at a point in a fluid. (moving or stationary)

Cartesian coordinate system

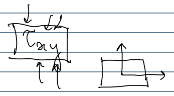
stress μ $\dot{\gamma}$

$$\tau_{xy} = \tau_{yx} = \mu\left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}\right)$$

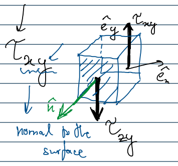
$$\tau_{yz} = \tau_{zy} = \mu\left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y}\right)$$

$$\tau_{zx} = \tau_{xz} = \mu\left(\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z}\right)$$

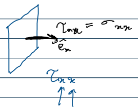
Laminar flow ✓



shear stress



$\tau_{zy} = \tau_{yz}$



Normal stress: $\sigma = \tau_{ii}$ (based on Hooke's law)

$\tau_{ij} = \tau_{ji}$ } Always true (angular momentum conservation)

normal part of force

pressure

$$\left(2\mu\frac{\partial v_x}{\partial x} - p\right)$$

$$\sigma_{xx} = \mu\left(2\frac{\partial v_x}{\partial x} - \frac{2}{3}\cancel{(\nabla\cdot\underline{v})}^{0}\right) - p$$

$$\sigma_{yy} = \mu\left(2\frac{\partial v_y}{\partial y} - \frac{2}{3}\cancel{(\nabla\cdot\underline{v})}^{0}\right) - p$$

$$\sigma_{zz} = \mu\left(2\frac{\partial v_z}{\partial z} - \frac{2}{3}\cancel{(\nabla\cdot\underline{v})}^{0}\right) - p$$

$(\nabla\cdot\underline{v}) = 0 \rightarrow$ Incompressible

Newtonian fluid

$$\boxed{\underline{\underline{\Pi}} = -p\underline{\underline{I}}}$$

Fluid Stress → Isotropic [same mag. of stress exerted on any surface & stress acts ~ Normal stress — Independent of surface orientation & always normal to any surface

→ Anisotropic (shear stress → different values of stress on different surfaces) } $\underline{\underline{\tau}}$

symmetric; 6 components

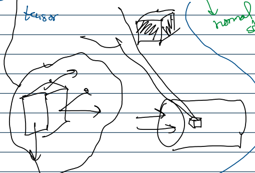
stress shear normal

$$\boxed{\underline{\underline{\Pi}} = \underline{\underline{\tau}} - p\underline{\underline{I}}}$$

P → Normal stress, Shear stress } $\underline{\underline{\Pi}}$ (tensor) $= \begin{pmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{pmatrix}$

surface surface

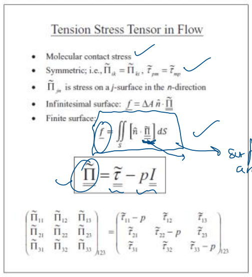
normal stress



$$-p\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{pmatrix} + \begin{pmatrix} 0 & -1 & 4 \\ -1 & 0 & 0 \\ 4 & 0 & 0 \end{pmatrix}$$

normal shear

$\underline{\underline{\Pi}}$



Tension Stress Tensor in Flow

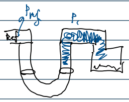
- Molecular contact stress ✓
- Symmetric; i.e., $\tilde{\Pi}_{ik} = \tilde{\Pi}_{ki}$, $\tilde{\tau}_{pm} = \tilde{\tau}_{mp}$ ✓
- $\tilde{\Pi}_{jn}$ is stress on a j-surface in the n-direction
- Infinitesimal surface: $\underline{f} = \Delta A\, \hat{n} \cdot \underline{\underline{\tilde{\Pi}}}$
- Finite surface:

$$\boxed{\underline{f} = \iint_S \left[\hat{n} \cdot \underline{\underline{\tilde{\Pi}}}\right] dS}$$ ✓ → surface analysis

$$\boxed{\underline{\underline{\tilde{\Pi}}} = \underline{\underline{\tilde{\tau}}} - p\underline{\underline{I}}}$$

$$\begin{pmatrix} \tilde{\Pi}_{11} & \tilde{\Pi}_{12} & \tilde{\Pi}_{13} \\ \tilde{\Pi}_{21} & \tilde{\Pi}_{22} & \tilde{\Pi}_{23} \\ \tilde{\Pi}_{31} & \tilde{\Pi}_{32} & \tilde{\Pi}_{33} \end{pmatrix}_{123} = \begin{pmatrix} \tilde{\tau}_{11} - p & \tilde{\tau}_{12} & \tilde{\tau}_{13} \\ \tilde{\tau}_{21} & \tilde{\tau}_{22} - p & \tilde{\tau}_{23} \\ \tilde{\tau}_{31} & \tilde{\tau}_{32} & \tilde{\tau}_{33} - p \end{pmatrix}_{123}$$

Figure 4.52 The stress tensor in matrix form can be used to calculate the tension on a surface in a flow.



Manometer Review

Pascal's principle : Pressure exerted on a confined liquid is transmitted equally to every part of the liquid & to the walls of the container } Idea of incompressibility

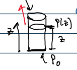
↓

applicable to confined fluids at rest

Manometer → device to accurately measure pressure differences ✓

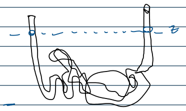
Momentum balance (RTT) ✓ → $\boxed{\frac{dp}{dz} = -\rho g}$ $\xrightarrow{\text{Integrate}}$ $\boxed{p(z) = -\rho g z + P_0}$ Pressure in a stationary fluid at elevation z (upward)

i.e. only dependent on the vertical height! (Not shape)



Pressures at same elevation (of same fluid) are eq

$p = f(z)$ ✓



PROBLEM: 4.16 A manometer is configured as shown in Figure 4.67. A heavy fluid (Fluid B) has been placed in the bottom of the manometer. A light fluid (Fluid A) has been added to the left side only. The left side of the manometer has been connected to a process stream in which water (25^oC, Fluid C) is flowing, while the right side of the manometer is open to the atmosphere (Fluid D). The manometer is being used to measure the pressure difference $p_1 - p_2$. The density of the two fluids in the manometer are $1.75\ g/cm^3$ and $13.6\ g/cm^3$. What is the pressure difference $p_1 - p_2$ in terms of fluid heights and fluid densities? For $h_1 = 2.3\ cm$, $h_2 = 2.3\ cm$, $h_3 = 1.0\ cm$, what is the pressure difference in psi?

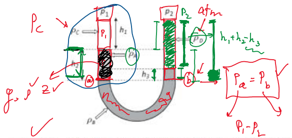
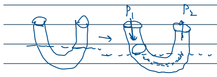


Figure 4.67 A schematic of a manometer that contains two different measurement fluids (Problem 16).

+ Pascals → incomp ✓



+ $p(z) = -\rho g z + P_0$ ✓

$P_{bottom} = P_{top} + \rho g z$

$P_a = P_B$ ✓ $P_{bot} = P_{top} + \rho g z$

P_{top} ↓ P_1

$\rho g z$ ↓ fluid 1 + fluid 2, ρ ρ

$\rho_A g \cancel{(h_1} + h_2)$

$$P_a = \underbrace{\rho_A g h_2 + \rho_C g h_1}_{} + P_1$$

$$P_b = \rho_B g h_3 + \rho_D g (h_1 + h_2 - h_3) + P_2$$

$$P_a = P_B$$

$$\Rightarrow P_1 + \rho_C g h_1 + \rho_A g h_2 = P_2 + \rho_D g h_1 + \rho_D g h_2 - \rho_D g h_3 + \rho_B h_3$$

$$\Rightarrow \boxed{(P_1 - P_2) = (\rho_D - \rho_C) g h_1 + (\rho_D - \rho_A) g h_2 + (\rho_B - \rho_D) g h_3}$$

PROBLEM: 5.35 We solved the incline problem using a coordinate system that sits on the solid surface (Figure 5.12). Here, choose instead to solve the same problem with a coordinate system in which the coordinate position $x = 0$ is located at the free surface (i.e., the top of the film) (Figure 5.42). What is the gravity acceleration vector in this coordinate system? What are the flow boundary conditions? Are there any advantages or disadvantages to this choice of coordinate system? Discuss your observations.